Web-Based Library Management System

Project Proposal Report

SE2030 – Software Engineering

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2. Introduction

2.1 Project Overview

The Web-Based Library Management System is designed to improve and automate the operations of a university library. Many libraries still rely on manual processes or outdated systems to manage book records, borrowing transactions, and user information. These methods can lead to errors, delays, and difficulty in tracking important data such as overdue books and fines.

The proposed system aims to provide a centralized web-based platform that allows librarians and users to manage and access library services efficiently. The system will store book records digitally, support issuing and returning of books, and provide real-time updates about book availability. By introducing an automated system, the library can improve operational efficiency and provide better service to its users.

This project focuses on designing a system that simplifies the management of library resources while ensuring accurate data handling and easy access for different stakeholders.

2.2 Objectives of the System

The main objective of the Web-Based Library Management System is to create an efficient and reliable platform for managing library operations.

The specific objectives include:

- Automating the management of book records in the library.
- Simplifying the process of issuing and returning books.
- Providing real-time information about book availability.
- Automatically calculating fines for overdue books.
- Generating reports to help library management monitor operations.
- Improving overall efficiency and accuracy in library management.

Through these objectives, the system aims to reduce manual work and improve the overall user experience for both library staff and students.

2.3 Target Users / Stakeholders

The proposed system will serve several types of users who interact with the library. Each stakeholder has different responsibilities and system needs.

Library Manager – Responsible for monitoring library operations and reviewing reports about book usage and overdue records.

Senior Librarian – Handles daily activities such as issuing and returning books and managing transactions.

Assistant Librarian – Maintains book records and updates the library database.

Student Services Officer – Handles student-related issues such as fine inquiries and borrowing history.

Senior Lecturer – Uses the library mainly for academic research and requires easy access to books and availability information.

Undergraduate Students – The main users of the library who search, borrow, and return books for academic purposes.

These stakeholders will interact with the system based on their roles and responsibilities.

2.4 Scope of the System

The scope of the Web-Based Library Management System includes the main functions required to manage library resources and user interactions.

The system will allow librarians to manage book records, issue and return books, and track overdue items. It will also provide users with the ability to search for books and check their availability. Additionally, the system will support automatic fine calculations and generate reports for management.

The system will be accessible through a web interface and will maintain a centralized database for storing all library-related information.

2.5 Limitations of the System

Although the proposed system will improve library operations, certain limitations may exist.

The system will require a stable internet connection to access the web platform. It will also be limited to authorized users within the university environment. Since the project is developed within a semester timeline, advanced features such as mobile applications or external library integrations may not be included in the initial version.

Despite these limitations, the system will provide essential functionality to improve library management and user experience.

3. System Overview Diagram

The System Overview Diagram illustrates the high-level structure of the Online Web-Based Library Management System and shows how different users interact with the system. The system is designed as a centralized web application that can be accessed through a web browser.

At the center of the system is the Library Management System platform, which processes all operations related to managing users, books, borrowing records, reservations, returns, and fines. All data is stored in a central database that maintains accurate and updated information.

The system includes several main modules that handle specific functions. The User Management Module manages user accounts, allowing administrators to register new users, update user information, view user details, and remove users when necessary. The Book Management Module maintains the library's book database by allowing librarians to add new books, update book details, view available books, and remove outdated records.

The Borrowing Management Module allows users to borrow books and keeps track of all borrowing transactions. If a user needs additional time, the borrowing period can be extended through the system. The Return Management Module records returned books and updates their availability in the system.

If a book is currently unavailable, the Reservation Management Module allows users to reserve the book and be notified when it becomes available. The Fine Management Module handles overdue fines by calculating penalties based on the number of late days and maintaining fine payment records.

Overall, the system integrates these modules to provide an efficient and organized platform for managing library operations. The web-based design ensures that authorized users such as librarians, students, and lecturers can access the system easily and perform their tasks through a centralized interface.

Online Web-Based Library Management System

4. Functional Requirements

The Online Web-Based Library Management System will support several core functions that allow library staff and users to manage library activities efficiently. These functions are designed based on the needs of the system users and the operations performed in a typical library environment.

1. User Management

The system must allow administrators or librarians to manage user accounts. This includes registering new users, viewing user details, updating user information, and removing users from the system when necessary. This function ensures proper management of students, lecturers, and staff who use the library system.

2. Book Management

The system must support managing book records in the library. Librarians should be able to add new books, view available books, update book details such as title or author, and remove books that are no longer available. This ensures that the library database remains accurate and up to date.

3. Borrowing Management

The system must allow users to borrow books from the library. It will record borrowing transactions, display currently borrowed books, allow extension of borrowing periods when necessary, and cancel borrowing records if needed. This function helps track all borrowing activities.

4. Return Management

The system must record book return transactions. Librarians can view return records, update return status, and remove incorrect return records if required. This ensures accurate tracking of returned books and updates the availability of books in the system.

5. Reservation Management

The system must allow users to reserve books that are currently unavailable. Users can create reservations, view existing reservations, modify reservation details, or cancel reservations when they are no longer needed. This feature ensures fair access to high-demand books.

6. Fine Management

The system must manage fines for overdue books. Librarians can add fines, view fine records, update payment status, and remove incorrect fine entries. The system will automatically calculate fines based on the number of late days.

Fine Calculation Formula:

Fine = Late Days × Fine per Day

Example:
Due Date = 14 June
Return Date = 16 June

Late Days = 2
Fine per Day = Rs.10

Total Fine = Rs.20

5. Non-Functional Requirements

Non-functional requirements describe the quality attributes of the system and how well the system performs its functions. These requirements focus on system performance, security, usability, reliability, and scalability.

1. Performance

The system should respond quickly to user requests. Common operations such as searching for books, borrowing records, and updating book information should be processed within a few seconds. The system should support multiple users accessing the platform simultaneously without significant delays.

2. Security

The system must ensure secure access to sensitive data. User authentication will be required through a secure login system. Role-based access control will be implemented so that different users (administrators, librarians, students, and lecturers) can only access the features relevant to their roles. Additionally, user information and system data must be protected from unauthorized access.

3. Usability

The system should provide a user-friendly interface that is easy to understand and navigate. Users should be able to perform tasks such as searching books, borrowing books, or managing records without requiring technical knowledge. Clear menus, simple forms, and intuitive navigation will improve the user experience.

4. Reliability

The system should operate consistently without frequent failures. Data such as book records, borrowing information, and fine details must be stored accurately in the database. Regular data backups should be maintained to prevent data loss.

5. Scalability

The system should be capable of handling future growth. As the number of users, books, and transactions increases, the system should continue to perform efficiently without affecting overall performance.

6. Availability

The web-based system should be available to authorized users whenever needed. Users should be able to access the system through the internet using a web browser, ensuring convenient access to library services.

6. Major Stakeholders / Users

The Online Web-Based Library Management System will be used by several stakeholders who interact with the system to perform different tasks related to library operations. Each stakeholder has specific responsibilities and access rights within the system.

1. Administrator

The administrator is responsible for managing the overall system. This role includes controlling system settings, managing user accounts, and ensuring that the system operates correctly. Administrators have full access to system functions such as user management, book management, and monitoring system activities.

2. Librarian

Librarians are responsible for managing the daily operations of the library. They use the system to add and update book records, manage borrowing and return transactions, and handle reservations made by users. Librarians also assist users in locating books and maintaining accurate library records.

3. Library Staff

Library staff support the librarian in managing the library system. They may help record book returns, update borrowing information, and assist in maintaining the database. Staff members have limited access to certain system features depending on their responsibilities.

4. Undergraduate Students

Undergraduate students are the primary users of the library system. They use the system to search for books, borrow books, reserve books that are currently unavailable, and view information about their borrowed books and fines.

5. Lecturers

Lecturers use the library system to search for academic resources required for teaching and research. They can borrow books, reserve books when necessary, and monitor their borrowing records through the system.

6. System Maintenance / IT Support

IT support personnel are responsible for maintaining the technical aspects of the system. Their responsibilities include system maintenance, database management, security monitoring, and resolving technical issues to ensure the system runs smoothly.

7. Six Major Functions

The Online Web-Based Library Management System will consist of six major functions. Each function is assigned to one group member and is responsible for managing a specific part of the system. These functions support the core operations of the library and improve the efficiency of managing books and users.

7.1 User Management

Assigned Member: IT21077524 – Pathirana A.P.C.E

Who will use it:
Administrators and librarians.

What it does:
The User Management function allows administrators to manage user accounts in the system. This includes registering new users, viewing user details, updating user information, and removing users when necessary. This module helps maintain accurate records of students, lecturers, and staff who use the library system.

Expected Outcome:
The system will maintain a well-organized user database, allowing administrators to easily manage user accounts and control access to the system.

7.2 Book Management

Assigned Member: IT21054372 – Samarasinghe P.D.P

Who will use it:
Librarians and library staff.

What it does:
The Book Management function allows librarians to manage book records in the library system. Librarians can add new books, view available books, update book information such as title or author, and remove books that are no longer available.

Expected Outcome:
The system will maintain an accurate and up-to-date library database that allows users to easily find and access book information.

7.3 Borrowing Management

Assigned Member: IT21158872 – Vimukthi W.M.B.V

Who will use it:
Librarians and students.

What it does:
The Borrowing Management function records when a user borrows a book from the library. The system stores details of the borrowing transaction, allows users to view borrowed books, and enables librarians to extend borrowing periods or cancel borrowing records if necessary.

Expected Outcome:
The system will accurately track all borrowing transactions and ensure that book availability is updated correctly.

7.4 Return Management

Assigned Member: IT21047756 – Weerasekara W.B.M.K.D

Who will use it:
Librarians and library staff.

What it does:
The Return Management function records when borrowed books are returned to the library. The system updates the return status, maintains return records, and updates the availability of the book so that it can be borrowed by other users.

Expected Outcome:
The system will maintain accurate return records and ensure that books become available in the system once they are returned.

7.5 Reservation Management

Assigned Member: IT21060144 – Rathnayake R M I

Who will use it:
Students and lecturers.

What it does:
The Reservation Management function allows users to reserve books that are currently unavailable. Users can create reservations, view reservation records, modify reservation details, or cancel reservations if they are no longer required.

Expected Outcome:
The system will allow users to secure access to books that are currently borrowed by others, improving fairness and availability of library resources.

7.6 Fine Management

Assigned Member: IT21101274 – Dolawatta T.Y.R

Who will use it:
Librarians and student service officers.

What it does:
The Fine Management function manages fines for overdue books. The system calculates fines automatically based on the number of late days and the defined fine rate per day. Librarians can view fine records, update payment status, and remove incorrect fine entries.

Expected Outcome:
The system will ensure accurate fine calculations and maintain proper records of fine payments, helping the library manage overdue penalties efficiently.

8. Minor Functions

In addition to the major functions of the Online Web-Based Library Management System, several minor or supportive functions are required to ensure smooth operation of the system.

These functions help users interact with the system easily and maintain proper system functionality.

1. User Login

The system will provide a secure login feature that allows authorized users to access the system using a username and password. Different user roles such as administrators, librarians, and students will have different levels of access to system features.

2. Logout Function

Users will be able to securely log out of the system after completing their tasks. This helps prevent unauthorized access to the system, especially when the system is used on shared computers.

3. Password Reset

The system will provide a password reset option for users who forget their login credentials. Users will be able to reset their passwords through a secure verification process.

4. Profile Management

Users will be able to view and update their personal information such as name, email, and contact details through their profile settings.

5. Notifications

The system may provide notifications to users regarding important activities such as due dates for borrowed books, reservation confirmations, or overdue fine alerts.

6. Search Filtering

The system will allow users to filter search results when looking for books. Users may search books by title, author, or category to quickly locate the required resources.

9. System Limitations / Constraints

The Online Web-Based Library Management System is designed to improve the efficiency of managing library resources. However, certain limitations and constraints may affect how the system operates.

1. Internet Dependency

Since the system is web-based, users must have a stable internet connection to access the platform. If the internet connection is slow or unavailable, users may experience delays or may not be able to access the system.

2. Platform Limitations

The system will primarily be designed to run on web browsers such as Google Chrome, Microsoft Edge, or Mozilla Firefox. Compatibility issues may occur if the system is accessed through unsupported or outdated browsers.

3. Device Requirements

The system is mainly intended to be used on desktop or laptop computers within the library or university environment. While the system may still be accessible through mobile devices, some features may not function optimally on smaller screens.

4. User Access Restrictions

Only authorized users such as administrators, librarians, lecturers, and registered students will be able to access the system. Unauthorized users will not be able to use the system features.

5. Development Time Constraints

Since the project is developed within a limited academic semester, only the core features of the library management system will be implemented. Advanced features such as mobile applications, external library integrations, or advanced analytics may not be included in the initial version.

6. Data Accuracy Assumptions

The system assumes that users such as librarians and administrators will enter correct and accurate data when managing books, borrowing records, and user information. Incorrect data entry may affect the accuracy of system records.

10. Simple Timeline

The following timeline shows the planned activities for developing the Online Web-Based Library Management System from Week 3 to Week 14. Each week includes the key tasks that the group intends to complete during the project development process.

Week	Planned Activities
Week 3	Submit project proposal and finalize project topic.

Week	Planned Activities
Week 4	Gather system requirements and identify stakeholders.
Week 5	Create system overview diagrams and finalize functional requirements.
Week 6	Design system architecture and database structure.
Week 7	Start development of User Management and Book Management modules.
Week 8	Develop Borrowing Management and Return Management modules.
Week 9	Implement Reservation Management functionality.
Week 10	Develop Fine Management module and fine calculation logic.
Week 11	Integrate all modules and test system functionality.
Week 12	Perform system testing and fix identified errors.
Week 13	Prepare final documentation and presentation materials.
Week 14	Final system submission and project presentation.

11. Conclusion

The proposed Online Web-Based Library Management System aims to improve the efficiency and organization of library operations by replacing manual processes with a centralized digital platform. The system will allow librarians, students, and lecturers to interact with the library more effectively through features such as user management, book management, borrowing and return tracking, reservations, and fine management.

By implementing this system, the library will be able to maintain accurate records of books and users while reducing errors that often occur in manual systems. The system will also provide faster access to information such as book availability, borrowing history, and overdue fines. These improvements will enhance the overall user experience and help library staff manage resources more efficiently.

The project focuses on developing a simple, reliable, and user-friendly system that meets the core needs of a modern library. By dividing the system into six main modules, each group member is responsible for developing and managing a specific function. This approach allows the team to work collaboratively while ensuring that all major parts of the system are completed effectively.

Our team plans to follow the proposed timeline and development plan to successfully complete the project within the semester. Through proper requirement analysis, system design, development, and testing, we aim to deliver a functional and efficient library management system that can support the needs of both library staff and users.

Overall, the Online Web-Based Library Management System provides a practical solution for improving library operations and demonstrates how technology can be used to enhance resource management and user accessibility in a university environment.